

Current algorithms for genus-two Ramond blocks

Ordinary theta sewing and recovery with a fermion insertion

Implementation notes

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These notes describe the two production algorithms in C++/: the ordinary NS–R–R theta block for $\eta\eta' = 1$, and the $\Theta v_{1/2}$ construction for $\eta\eta' = -1$. The output in both cases is the physical superconformal block $\mathbb{F}_f^{(\eta,\eta')}$, including its dependence on the three spin lifts. The shared ingredients are branching Ward identities, forward central-charge recursion for the two ordinary Virasoro factors, the Schottky vacuum, direct fermion sewing, and sector-restricted convolution. Only the current algorithms and their implemented optimizations are described. The mathematical statements concern generic parameters; numerical solves retain rank and residual checks. No arbitrary-genus frontend is claimed for this code.

The computation proceeds in five steps: construct and reuse the finite action data; evaluate and store the outer branching recurrence and, for the insertion, the middle recurrence; evaluate the reduced Virasoro factors at branch-dependent cutoffs; assemble the enlarged coefficients and divide by the direct fermion factor in the appropriate parity sector; restore the common Schottky vacuum squared. For the inserted graph only equal *total* levels on its two middle segments are needed. Section 5 proves why this restriction closes under recovery and gives the exact smaller domains needed inside the two Virasoro factors.

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1 Conventions and the two recovery problems

The two spheres have punctures $(\infty, 1, 0)$ with sectors (NS, R, R). Edges 1, 2, 3 have plumbing parameters q_1, q_2, q_3 and spin lifts $\eta_1, \eta_2, \eta_3 \in \{\pm 1\}$. The vertex signs η, η' are distinct from these spin lifts; $f \in \{0, 1\}$ labels the three-point form, and p_1 is the intrinsic parity of the NS primary. We use the conventions of `Human Notes/SCblock.tex`, with the contour transport specified below. All contractions are bilinear BPZ contractions, without complex conjugating the vertex coefficients. Hermitian norms appear only in numerical conditioning. Total level N means retaining $q_1^{a/2} q_2^{l_2} q_3^{l_3}$ with $a, l_2, l_3 \geq 0$ integers and $a + 2l_2 + 2l_3 \leq 2N$. The alternative independent-edge cutoff retains $0 \leq a \leq 2N$ and $0 \leq l_2, l_3 \leq N$ without a bound on their sum. The command-line choices are `-truncation total` (the default) and `-truncation per-edge`; the latter applies the same numerical limit N independently to all three edges. The fresh timings below use this second choice at $N = 10$, including the corner $q_1^{10} q_2^{10} q_3^{10}$.

Write

$$Q = b + b^{-1}, \quad c = \frac{3}{2} + 3Q^2, \quad h_1 = \frac{Q^2/4 - P_1^2}{2}, \quad \beta_j = P_j/\sqrt{2} \quad (j = 2, 3). \quad (1.1)$$

The physical Ramond weights are $c/24 - \beta_j^2$. The implementation takes real $b \neq 0, \pm 1$ and permits complex momenta. Genericity excludes singular embedding norms, unresolved Virasoro pole collisions, and points where neither branching Ward identity determines the required coefficient. A vanishing prefactor in the first identity alone is handled by the second identity. Genuine singular limits require a separate meromorphic-continuation prescription.

The physical Ramond ground conventions and three-point initial data are

$$G_0 w^\pm = i\beta e^{\mp i\pi/4} w^\mp, \quad (\langle w^\alpha, w^\gamma \rangle) = \text{diag}(1, i), \quad (1.2)$$

	++	--	+-	-+
$\rho_0^{(\eta)}(\phi, w^\alpha, w^\gamma)$	1	η	0	0
$\rho_1^{(\eta)}(\phi, w^\alpha, w^\gamma)$	0	0	1	$i\eta$

(1.3)

The auxiliary grounds satisfy $\psi_0 u^b = u^{1-b}/\sqrt{2}$ and have metric $\text{diag}(1, -1)$. Tensor products are ordered auxiliary first; a physical odd operator therefore acquires the auxiliary state's parity.

For parity monomials, the sewing convolution is

$$\prod_i \eta_i^{\epsilon_i} \star_R \prod_i \eta_i^{\delta_i} = (-1)^{\sum_{i < j} (\epsilon_i \delta_j + \delta_i \epsilon_j)} \prod_i \eta_i^{\epsilon_i + \delta_i}. \quad (1.4)$$

Parity exponents are reduced modulo two; plumbing exponents add ordinarily. The cocycle is the difference between the quadratic theta sign at total parity and the two separate quadratic signs. It is symmetric and bilinear, so this product is commutative and associative.

The physical block obeys, coefficient by coefficient,

$$(\eta_2 \eta_3) \star_R \mathbb{F}_f^{(\eta, \eta')} = \eta \eta' \mathbb{F}_f^{(\eta, \eta')}. \quad (1.5)$$

One way to derive this is to change both Ramond summation bases by

$$\mathbb{L}_{-B} w^+ \mapsto (-1)^{\#G(B)} e^{-i\pi/4} \mathbb{L}_{-B} w^-, \quad \mathbb{L}_{-B} w^- \mapsto -(-1)^{\#G(B)} e^{i\pi/4} \mathbb{L}_{-B} w^+. \quad (1.6)$$

Here $\#G(B)$ counts supercurrent modes. This odd intertwiner preserves the ground metric $\text{diag}(1, i)$ and supercommutes with the SCA. Applying the map componentwise to both Ramond arguments multiplies a physical three-point form by $-i\eta(-1)^{\epsilon_2}$. The graded tensor action supplies another $(-1)^{\epsilon_2}$ and hence has factor $-i\eta$. In the sewn pair of vertices the componentwise parity factors cancel, giving $-\eta\eta'$. Flipping the two parity bits changes the theta sign by $(-1)^{\epsilon_2 + \epsilon_3 + 1}$. Thus the coefficients at the flipped and original parities differ by $\eta\eta'(-1)^{\epsilon_2 + \epsilon_3}$, which is exactly (1.5) by (1.4).

Ordinary sewing factorizes as $\widehat{\mathbb{F}}_f^{(\eta, \eta')} = \mathbb{F}_f^{(\eta, \eta')} \star_R \mathbb{F}_F$. The uninserted auxiliary block satisfies $(\eta_2 \eta_3) \star_R \mathbb{F}_F = \mathbb{F}_F$ and has constant $1 + \eta_2 \eta_3$. Its convolution with the negative sector vanishes in one line:

$$\mathbb{F}_F \star_R \mathbb{F}_f^{(\eta, -\eta)} = \mathbb{F}_F \star_R (\eta_2 \eta_3) \star_R \mathbb{F}_f^{(\eta, -\eta)} = -\mathbb{F}_F \star_R \mathbb{F}_f^{(\eta, -\eta)} = 0. \quad (1.7)$$

This is cancellation of the two Ramond ground contributions, equivalently an unsaturated auxiliary zero mode in the corresponding loop trace. It is not a vanishing statement about the physical block. For the positive sector, the same auxiliary constant acts as the scalar 2. For the negative sector we instead construct an auxiliary constant $Q(1 - \eta_2 \eta_3)/\sqrt{2}$, which acts as $\sqrt{2}Q$. Neither constant is a unit in the entire parity algebra.

2 Branching conventions and the enlarged blocks

The two commuting Virasoro algebras are generated by

$$L_m^{(1)} = \frac{b^{-1}L_m - (b^{-1} + 2b)L_m^F + U_m}{b^{-1} - b}, \quad L_m^{(2)} = \frac{bL_m - (b + 2b^{-1})L_m^F + U_m}{b - b^{-1}}, \quad (2.1)$$

$$L_m^F = \frac{1}{2} \sum_r r : \psi_{m-r} \psi_r : + \frac{1-2\nu}{16} \delta_{m0}, \quad U_m = \sum_r \psi_{m-r} G_r. \quad (2.2)$$

Here $r \in \mathbb{Z} + \nu$, with $\nu = 1/2$ in NS and $\nu = 0$ in R. We use $\psi_r^\dagger = -\psi_{-r}$, $L_m^\dagger = L_{-m}$, $G_r^\dagger = G_{-r}$. The weights and central charges are

$$h_{j,n}^{(1)} = \frac{Q^2/4 - (P_j + 2nb)^2}{2(1 - b^2)}, \quad c^{(1)} = 1 + \frac{3Q^2}{1 - b^2}, \quad (2.3)$$

$$h_{j,n}^{(2)} = \frac{Q^2/4 - (P_j + 2n/b)^2}{2(1 - b^{-2})}, \quad c^{(2)} = 1 + \frac{3Q^2}{1 - b^{-2}}. \quad (2.4)$$

The branch labels are $n_1 \in \frac{1}{2}\mathbb{Z}$ and $n_2, n_3 \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}$. Their levels above the enlarged ground states are $2n_1^2$ and $2n_j^2 - 1/8$, respectively. In particular the Ramond subtraction is $1/8$, not the $1/16$ shift between an enlarged weight and the physical Ramond weight.

2.1 Primary normalization and the vertex frame

Put $\chi_r(P) = \psi_r - i\eta_r(P)$, where η_r is the physical free-field fermion. For $n > 0$ the NS primary is

$$v_n(P) = 2^{-2n} \ell(Q + 2P, 4n) \chi_{-1/2}(P) \chi_{-3/2}(P) \cdots \chi_{-(4n-1)/2}(P) \phi(P), \quad v_0 = \phi. \quad (2.5)$$

This is the raw operator order used in C++. Reversing the $2n$ fermions to the order in the human notes multiplies the primary by $(-1)^{n(2n-1)}$. This phase changes raw vertices and action coefficients consistently and cancels between the two sewn vertices. For Ramond $n > 0$, put $M = 2n - 1/2$ and let t be the residue of $M + 1$ modulo two. The two primaries are

$$v_n^t = \chi_0(P) \chi_{-1}(P) \cdots \chi_{-M}(P) (u^0 \otimes w^+), \quad (2.6)$$

$$v_n^{1-t} = \chi_0(P) \chi_{-1}(P) \cdots \chi_{-M}(P) \chi_0(-P) (u^0 \otimes w^+), \quad \chi_0(-P) = \psi_0 + i\eta_0(P). \quad (2.7)$$

Negative labels use simultaneous $n \mapsto -n$, $P \mapsto -P$; on physical Ramond grounds the intertwiner is $\text{diag}(1, -1)$. This fixes the raw normalization, including the two parity copies.

For positive integer m ,

$$\ell(x, m) = 2^{(m \bmod 2)/8} \prod_{\substack{r, s \geq 0, \ r+s < m \\ r+s \equiv m \pmod{2}}} (x + rb + s/b), \quad \ell(x, 0) = 1. \quad (2.8)$$

For $n > 0$, the squared BPZ norms used in assembly are

$$\|v_n\|^2 = (-1)^{2n} 2^{-2n} \ell(2P, 4n) \ell(Q + 2P, 4n), \quad (2.9)$$

$$\|v_n^0\|^2 = 2^{2\lfloor M/2 \rfloor + 1} \frac{\ell(2P, 4n)}{\ell(Q + 2P, 4n)}, \quad \|v_n^1\|^2 = -2^{2\lceil M/2 \rceil} \frac{\ell(2P, 4n)}{\ell(Q + 2P, 4n)}. \quad (2.10)$$

Negative labels again use reflection. Using squared norms avoids numerical choices of square roots for normalized branching coefficients.

The code uses physical ground coordinates $(w^+, e^{3\pi i/4} w^-)$, whose metric is $(1, 1)$, and the Ward sign $\eta_{\text{native}} = (-1)^f \eta$. After transporting these coordinates, the implemented enlarged vertex is the following tensor-product form. Let A, B be the physical descendant words in the first two slots and A, B, C the auxiliary words; α is the physical second-slot ground parity:

$$\begin{aligned} \hat{\rho}_f^{(\eta)} = & (-i)^{\#G(A) \bmod 2} (-1)^{f(\#G(A) + \#G(B) + \alpha) + \#G(A) \#A} \\ & \times (-1)^{(\#G(B) + \alpha + p_1)(\#C + c)} \rho_F \rho_f^{(\eta)}. \end{aligned} \quad (2.11)$$

The arguments are in the order (NS, R, R), with auxiliary third ground u^c . The physical forms have the BPZ metric above and auxiliary ground values $\rho_F(\mathbf{1}, u^0, u^0) = 1$, $\rho_F(\mathbf{1}, u^1, u^1) = i$. Here the unhatted physical form uses the human ground and contour conventions specified above; the powers of $-i$ and the f -dependent sign implement its transport, rather than redefine its ground data. These are parts of the definition, not adjustable phases in the final sum: the two $(-i)^{\#G(A) \bmod 2}$ factors cancel the physical NS contour sign.

2.2 Ordinary theta block

All Virasoro blocks F below have leading coefficient one, with primary plumbing powers displayed separately. The ordinary enlarged block is

$$\begin{aligned} \hat{\mathbb{F}}_f^{(\eta, \eta')} = & \sum_{\substack{n_1 \in \frac{1}{2}\mathbb{Z}, \ n_2, n_3 \in \frac{1}{2}\mathbb{Z} + \frac{1}{4} \\ \alpha, \gamma \in \{0, 1\} \\ 2n_1 + \alpha + \gamma \equiv f \pmod{2}}} q_1^{2n_1^2} q_2^{2n_2^2 - 1/8} q_3^{2n_3^2 - 1/8} \eta_1^{2n_1 + p_1} \eta_2^\alpha \eta_3^\gamma \\ & \times (-1)^{2n_1 + (2n_1 + p_1)(\alpha + \gamma) + \alpha\gamma} \frac{\hat{\rho}_f^{(\eta)}(v_{1,n_1}^{p_1}, v_{2,n_2}^\alpha, v_{3,n_3}^\gamma) \hat{\rho}_f^{(\eta')}(v_{1,n_1}^{p_1}, v_{2,n_2}^\alpha, v_{3,n_3}^\gamma)}{\|v_{1,n_1}^{p_1}\|^2 \|v_{2,n_2}^\alpha\|^2 \|v_{3,n_3}^\gamma\|^2} \\ & \times \prod_{i=1}^2 F(h_{1,n_1}^{(i)}, h_{2,n_2}^{(i)}, h_{3,n_3}^{(i)}; c^{(i)} \mid q_1, q_2, q_3). \end{aligned} \quad (2.12)$$

The label ranges are those following (2.4). The fraction is the product of the two normalized branching coefficients $\mathbb{B}_f^{(\eta); p_1} \mathbb{B}_f^{(\eta'); p_1}$. For $\eta\eta' = 1$ this sum, divided by the ordinary auxiliary factor in its sector, determines the physical block.

2.3 Insertion for the negative Ramond loop

Split edge 2 by a sphere with coordinate z_3 :

$$(z_1 - 1)z_3 = \hat{q}_{2,1}, \quad (z_2 - 1)/z_3 = \hat{q}_{2,2}, \quad q_2 = \hat{q}_{2,1}\hat{q}_{2,2}. \quad (2.13)$$

Choose transported spin frames so that $\eta_2 = \hat{\eta}_{2,1}\hat{\eta}_{2,2}$. In the irreducible physical NS vacuum module, $G_{-1/2}\mathbf{1} = 0$, so $v_{1/2}(Q/2) = Q\psi_{-1/2}\mathbf{1}$. At $z_3 = 1$ insert the ordered operator $Q\Theta\psi(1)$, where

$$\Theta\Psi_{-B}u^b = (-1)^{\#B+b+1}\Psi_{-B}u^{1-b}. \quad (2.14)$$

It obeys $\Theta^2 = -1$, anticommutes with ψ_r and the physical G_r , and commutes with L_m and both $L_m^{(i)}$. Thus $\Theta\psi$ is even and acts only on the auxiliary factor. In particular

$$Q\Theta\psi_0\Psi_{-B}u^b = \frac{Q}{\sqrt{2}}(-1)^b\Psi_{-B}u^b. \quad (2.15)$$

The extra sign between Ramond grounds removes the cancellation in (1.7).

The inserted double-Virasoro field has weights

$$h_\psi^{(1)} = -\frac{1 + 2b^2}{2(1 - b^2)}, \quad h_\psi^{(2)} = \frac{b^2 + 2}{2(1 - b^2)}. \quad (2.16)$$

These have level-two degenerate Kac labels $(2, 1)$ in the first copy and $(1, 2)$ in the second, in the embedding convention. At common physical momentum their two fusion rules restrict the middle branches to $n' = n \pm 1/2$. Both halves carry the same enlarged parity α , because the field and Θ each reverse it. Let the punctured Virasoro block have ordered internal weights (h_1, h_L, h_R, h_3) and external weight h_ψ ; its outer vertices are (h_1, h_L, h_3) and (h_1, h_R, h_3) , and its middle vertex is

(h_R, h_ψ, h_L) at $(\infty, 1, 0)$. The enlarged block is

$$\begin{aligned}
\hat{\mathbb{F}}_f^{(\eta, \eta')}[\Theta v_{1/2}] = & \sum_{\substack{n_1 \in \frac{1}{2}\mathbb{Z}, \ n, n', n_3 \in \frac{1}{2}\mathbb{Z} + \frac{1}{4} \\ \alpha, \gamma \in \{0, 1\} \\ n' = n \pm 1/2, \ 2n_1 + \alpha + \gamma \equiv f \ (2)}} q_1^{2n_1^2} \hat{q}_{2,1}^{2n^2-1/8} \hat{q}_{2,2}^{2n'^2-1/8} q_3^{2n_3^2-1/8} \eta_1^{2n_1+p_1} \eta_2^\alpha \eta_3^\gamma \\
& \times (-1)^{2n_1+(2n_1+p_1)(\alpha+\gamma)+\alpha\gamma} \frac{\hat{\rho}_f^{(\eta)}(v_{1,n_1}^{p_1}, v_{2,n}^\alpha, v_{3,n_3}^\gamma) \hat{\rho}_f^{(\eta')}(v_{1,n_1}^{p_1}, v_{2,n'}^\alpha, v_{3,n_3}^\gamma)}{\|v_{1,n_1}^{p_1}\|^2 \|v_{2,n}^\alpha\|^2 \|v_{2,n'}^\alpha\|^2 \|v_{3,n_3}^\gamma\|^2} \\
& \times \langle v_{2,n'}^\alpha | Q \Theta \psi(1) | v_{2,n}^\alpha \rangle \prod_{i=1}^2 F(h_{1,n_1}^{(i)}, h_{2,n}^{(i)}, h_{2,n'}^{(i)}, h_{3,n_3}^{(i)}; h_\psi^{(i)}; c^{(i)} \mid q_1, \hat{q}_{2,1}, \hat{q}_{2,2}, q_3). \quad (2.17)
\end{aligned}$$

The additional inverse primary norm belongs to the additional cut. The numerical calculation evaluates only the shifted diagonal of this sewing expression, on the domains in Section 5. The middle matrix element is a branching coefficient: in normalized radial ordering it is $i\|v_{1/2}\|\mathbb{B}(n', 1/2, n; \alpha, 1-\alpha)$ times the two internal norm roots, with compatible roots satisfying $\Theta(v_n^\alpha/\|v_n^\alpha\|) = -iv_n^{1-\alpha}/\|v_n^{1-\alpha}\|$. Equation (2.17) uses raw matrix elements and is independent of that optional square-root notation. Here $\|v_{1/2}(Q/2)\|^2 = -Q^2$.

3 Computing and reusing the branching data

3.1 Finite action systems

The columns are constructed from oscillators with

$$[c_m, c_n] = m\delta_{m+n,0}, \quad \{\eta_r, \eta_s\} = \{\psi_r, \psi_s\} = \delta_{r+s,0}, \quad \{\eta_r, \psi_s\} = 0. \quad (3.1)$$

On the positive chart the physical free-field realization is

$$L_m = \frac{1}{2} \sum_{k \neq 0, m} c_k c_{m-k} + \frac{1}{2} \sum_r r \eta_{m-r} \eta_r + \frac{i}{2} (Qm - 2P) c_m \quad (m \neq 0), \quad (3.2)$$

$$G_r = \sum_{k \neq 0} c_k \eta_{r-k} + i(Qr - P) \eta_r, \quad (3.3)$$

$$L_0 = \frac{Q^2/4 - P^2}{2} + \frac{1 - 2\nu}{16} + \sum_{k > 0} c_{-k} c_k + \sum_{r > 0} r \eta_{-r} \eta_r. \quad (3.4)$$

Positive modes annihilate the ground states. On the internal Ramond physical coordinates, η_0 swaps the grounds with coefficient $1/\sqrt{2}$. Reflection replaces P by $-P$ and transports the ground as specified above. Only finitely many terms of each mode sum act nontrivially on a given state.

For positive labels the required action expansions are

$$L_1 v_n = \sum_{|A|+|B|=4n-3} \mathbf{V}_{1,n}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1} \quad (n \geq 1), \quad (3.5)$$

$$L_1 v_n^\alpha = \sum_{|A|+|B|=4n-3} \mathbb{V}_{1,n,\alpha}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}^\alpha \quad (n \geq 3/4), \quad (3.6)$$

$$L_{-1} v_n^\alpha = \sum_{i=1}^2 \mathbb{V}_{n,\alpha}^{(i)} L_{-1}^{(i)} v_n^\alpha + \sum_{|A|+|B|=4n-1} \mathbb{V}_{-1,n,\alpha}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}^\alpha \quad (n \geq 3/4). \quad (3.7)$$

At NS $n = 0, \pm 1/2$ and R $n = \pm 1/4$, L_1 annihilates the primary. At $n = 1/4$, the last line instead has the two same-primary level-one terms and a primary term $v_{-3/4}^\alpha$; reflection supplies $n = -1/4$. The code constructs this small boundary span directly. It does not extend the generic $n \mapsto n - 1$ formula across this boundary.

Expand the target on the left and every candidate descendant on the right in the free-field oscillator basis. Solve the resulting finite column-span problem for $\mathbf{V}$ or $\mathbb{V}$. The candidate supports in the displayed action expansions are the structural input from the branching identities; the implemented all-row residual checks test that the computed target lies in those supports at the requested parameters. They are not a proof of these identities at arbitrary level. The ordinary pipeline requires the NS L_1 and Ramond L_{-1} data. The insertion additionally requires Ramond L_1 data on edge 2. When the second outer Ward identity is needed, NS L_{-1} and Ramond L_0, L_1 actions are constructed lazily and stored. Here physical L_0 acts with the physical oscillator level, including the Ramond ground weight; it is not replaced by the enlarged conformal weight.

3.2 Outer branching from directly computed boundary values

First directly compute and store all allowed coefficients in the boundary box

$$n_1 = 0, \pm \frac{1}{2}, \quad n_2, n_3 = \pm \frac{1}{4}, \pm \frac{3}{4}. \quad (3.8)$$

This is the boundary prescription in `Human Notes/SCblock.tex`. The explicit free-field states are converted to small physical PBW states; their physical and auxiliary three-point forms follow from the Ward identities with (2.11). Only these initial values and low boundary conversions use physical PBW. No high-level physical Gram sewing enters.

For higher labels, use the physical conformal Ward identities

$$\hat{\rho}(L_1 v_{1,n_1}, v_{2,n_2}, v_{3,n_3}) = \hat{\rho}(v_{1,n_1}, L_{-1} v_{2,n_2}, v_{3,n_3}) + \hat{\rho}(v_{1,n_1}, v_{2,n_2}, L_{-1} v_{3,n_3}), \quad (3.9)$$

$$\begin{aligned} \hat{\rho}(v_{1,n_1}, L_1 v_{2,n_2}, v_{3,n_3}) &= \hat{\rho}(L_{-1} v_{1,n_1}, v_{2,n_2}, v_{3,n_3}) - \hat{\rho}(v_{1,n_1}, L_{-1} v_{2,n_2}, v_{3,n_3}) \\ &\quad - 2\hat{\rho}(v_{1,n_1}, L_0 v_{2,n_2}, v_{3,n_3}) - \hat{\rho}(v_{1,n_1}, v_{2,n_2}, L_1 v_{3,n_3}). \end{aligned} \quad (3.10)$$

Insert the mode-action decompositions and factor descendant forms into the raw primary coefficient and the two ordinary Virasoro three-point factors. The first identity usually determines a coefficient from stored lower-label values by a scalar division. If its prefactor vanishes, use the second identity. Store every computed coefficient immediately and reuse the same Ward coefficients for both requested vertex signs.

The actual dependency order increases

$$\lfloor |n_1| \rfloor + \lfloor |n_2| - \frac{1}{4} \rfloor + \lfloor |n_3| - \frac{1}{4} \rfloor. \quad (3.11)$$

The boundary action exchanges the pairs $\{1/4, -3/4\}$ and $\{-1/4, 3/4\}$ on a Ramond leg; the second identity may also exchange $\{1/2, -1/2\}$ on the NS leg. These exchanges preserve the displayed order. Group such coupled coefficients at fixed order and substitute every stored lower-order value first. Outside the directly seeded box, at most four unknowns remain in a group. Solve this local relation at the working precision, then continue upward. Thus the bulk update is scalar and the small boundary couplings never become a global Ward matrix. Every external dependency is checked to have smaller order and to be present in the cache.

Pivot tests use the contributions to the local prefactor and the other unknowns in its small group. Large already-known lower-label contributions are not included in this scaling. If the first identities

do not determine the group, append the corresponding second identities and retry locally. After each update, evaluate all used identities with the stored coefficients. An unresolved pivot or failed residual reports the particular labels, parities, and, for a residual failure, vertex sign.

Start from the primary triples required by the branch sum and close this set under the action-label changes. Retain $2n_1 + \alpha + \gamma \equiv f \pmod{2}$. For the inserted graph, initial candidates use the average of the two middle primary levels under a total cutoff; the stricter maximum-level condition in Section 5 is imposed during diagonal assembly. Under an independent-edge cutoff, both middle primary levels must already be at most N , and the other primary levels are bounded on their own edges. The closure may add lower-label coefficients needed by the Ward recurrence beyond these initial branch requests. The action stage still prepares its edge-label ranges before this closure. The ordinary pipeline requests one vertex sign; the inserted pipeline requests both, with the same local elimination applied to their two right-hand sides.

For a word $A = (a_1, \dots, a_k)$, put $s_j = \sum_{t>j} a_t$. The normalized single-leg descendant factors are the closed products

$$\rho(L_{-A}\nu_1, \nu_2, \nu_3) = \prod_j (h_1 + s_j + a_j h_2 - h_3), \quad (3.12)$$

$$\rho(\nu_1, L_{-A}\nu_2, \nu_3) = \prod_j (-1)^{a_j} (h_2 + s_j + a_j h_3 - h_1), \quad (3.13)$$

$$\rho(\nu_1, \nu_2, L_{-A}\nu_3) = \prod_j (h_3 + s_j + a_j h_2 - h_1). \quad (3.14)$$

These replace recursive three-point evaluation in the outer and middle Ward equations. Products with the same changed labels, slot, copy, and word are cached in the outer recurrence. The embedded weights are cached by leg, label, and Virasoro copy; empty descendant words return one immediately.

3.3 Middle recurrence

Since $[L_m, Q\Theta\psi(1)] = 0$ for the physical stress tensor,

$$\langle L_1 v_{n'}^\alpha | Q\Theta\psi(1) | v_n^\alpha \rangle = \langle v_{n'}^\alpha | Q\Theta\psi(1) | L_{-1} v_n^\alpha \rangle, \quad (3.15)$$

and the transposed identity also holds. For $n' = n + 1/2$ with $n \geq 1/4$, substitution gives

$$\begin{aligned} & \left[\sum_{i=1}^2 \mathbb{V}_{n,\alpha}^{(i)} (h_{2,n}^{(i)} + h_{\psi}^{(i)} - h_{2,n'}^{(i)}) \right] \langle v_{n'}^\alpha | Q\Theta\psi(1) | v_n^\alpha \rangle \\ &= \sum_{|A|+|B|=4n'-3} \mathbb{V}_{1,n',\alpha}^{AB} \rho(L_{-A}\nu_{h_{2,n'-1}^{(1)}}, \nu_{h_{\psi}^{(1)}}, \nu_{h_{2,n}^{(1)}}) \rho(L_{-B}\nu_{h_{2,n'-1}^{(2)}}, \nu_{h_{\psi}^{(2)}}, \nu_{h_{2,n}^{(2)}}) \langle v_{n'-1}^\alpha | Q\Theta\psi(1) | v_n^\alpha \rangle. \end{aligned} \quad (3.16)$$

The other-primary term of $L_{-1}v_n$ is separated from n' by $3/2$ and vanishes by fusion. For the reverse orientation use the transposed identity, with the descendant on the incoming leg. More generally apply L_1 to the endpoint of larger absolute label, using reflection for negative labels. Each step decreases that absolute label and reaches an oriented ground pair. Equation (2.15) fixes both orientations:

$$\langle v_{1/4}^0 | Q\Theta\psi | v_{-1/4}^0 \rangle = \sqrt{2}Q, \quad \langle v_{1/4}^1 | Q\Theta\psi | v_{-1/4}^1 \rangle = Q/\sqrt{2}. \quad (3.17)$$

The reversed pairs have the same values. The recurrence uses cached actions from the outer stage and stores each ordered pair and parity once.

3.4 Implemented reuse and numerical linear algebra

The following optimizations are part of the current C++ implementation.

1. **Reflection and parity transport.** Negative-label actions are computed on the positive chart at $-P$ and relabelled, avoiding a large Fock-to-PBW reflection matrix. Only parity-zero Ramond action systems are solved. For a term $n \rightarrow m$, its parity-one coefficient is multiplied by

$$2^{\{(-1)^{2|m|-1/2} - (-1)^{2|n|-1/2}\}/2}. \quad (3.18)$$

This follows by commuting the action through $\Theta v_n^0 = -2^{(-1)^{2|n|-1/2}/2} v_n^1$. It includes the factors $1/2$ and 2 at the $1/4 \leftrightarrow -3/4$ boundary.

2. **Interned sparse states.** Each oscillator occupation pattern has one integer identifier. Sparse linear combinations store identifiers and coefficients. Nonzero fermion occupations use bit masks; their reordering signs come from population counts. Physical-mode images ignore auxiliary spectators; auxiliary stress-tensor images ignore physical spectators. Spectators and graded signs are restored when the stored image is used. The fixed oscillator storage checks its mode bounds explicitly (64 positive fermion slots and boson modes $1, \dots, 63$).
3. **Commuting-copy scheduling.** Descendant words share stored suffix states and embedded-generator images. At each step the code selects the larger available leading mode from the two commuting Virasoro words. This changes intermediate work, not either word's internal order. It uses commutation of the two copies; it does not identify their different weights or numerical blocks.
4. **Paired actions and cache lifetimes.** Required L_{-1} and L_1 actions on edge 2 share primary and descendant construction before their temporary image/suffix caches are released. Solved actions, primaries, and interned states remain available to the run. No precomputed numerical file is loaded. The two physical momenta and their reflected charts remain distinct.
5. **Stored branching recurrence.** Each directly evaluated boundary value and each subsequently computed coefficient is retained. Ward rows, embedded weights, and single-leg descendant factors are reused. The two vertex signs share the same local elimination; its dimension is at most four.
6. **Scaled action systems.** Columns are normalized once. Native action solves use full-column-rank LAPACK least squares. Multiprecision action systems use pivoted QR of the transposed scaled matrix to select independent rows, then an LU factorization of that square restriction.
7. **Multiprecision refinement.** Double precision supplies row choices and an LU preconditioner only. Residuals are evaluated using the original multiprecision columns, correction equations use the reusable preconditioner, and corrections are accumulated at working precision. The final residual is checked on every original row, including unselected rows, using sparse entries. This obtains accurate coefficients without factoring a large dense MPC matrix.

4 Forward CCY recursion and the Schottky vacuum

Both Virasoro factors use the central-charge recursion of Cho, Collier and Yin [1]. Separate the universal vacuum factor F_{vac} and write $F = F_{\text{vac}} F_{\text{red}}$. At descendant multi-level ℓ ,

$$F_{\text{red},\ell}(c; \mathbf{h}) = G_\ell(\mathbf{h}) + \sum_e \sum_{\substack{r \geq 2 \\ s \geq 1}} \frac{R_{rs}^{(e)}(\mathbf{h})}{c - c_{rs}(h_e)} F_{\text{red},\ell - r s \mathbf{e}_e}(c_{rs}(h_e); \mathbf{h} + r s \mathbf{e}_e). \quad (4.1)$$

Only nonnegative remaining levels contribute; $\mathbf{e}_e$ is the unit vector on internal edge e , and G is the global $SL(2)$ coefficient. The external inserted weights remain fixed during all recursion steps, including steps that change the central charge.

4.1 Residues and their reusable factors

With $x = b_{rs}(h)^2$,

$$x = \frac{rs - 1 + 2h \pm \sqrt{(r-s)^2 + 4(rs-1)h + 4h^2}}{1 - r^2}, \quad c_{rs}(h) = 13 + 6(x + x^{-1}), \quad (4.2)$$

$$R_{rs}^{(e)} = -\partial_{h_e} c_{rs}(h_e) A_{rs} P_{rs}(h_a, h_b) P_{rs}(h_d, h_k), \quad (4.3)$$

$$A_{rs} = \frac{1}{2} \prod_{\substack{u=1-r,\dots,r; \ v=1-s,\dots,s \\ (u,v) \neq (0,0), (r,s)}} (ub_{rs} + v/b_{rs})^{-1}. \quad (4.4)$$

The two fusion polynomials use the spectator pairs at the two endpoints:

Null edge	Ordinary theta (both endpoints)	Inserted graph (two endpoints)
1	(h_3, h_2)	$(h_3, h_L), (h_3, h_R)$
2 or L	(h_1, h_3)	$(h_3, h_1), (h_R, h_\psi)$
R	—	$(h_3, h_1), (h_L, h_\psi)$
3	(h_1, h_2)	$(h_1, h_L), (h_1, h_R)$

The order of each pair specifies the sign at odd null level. For $\lambda_a^2 = x + 2 + x^{-1} - 4h_a$ the fusion polynomial is

$$P_{rs}(h_a, h_b) = \prod_{\substack{u=1-r, 3-r, \dots, r-1 \\ v=1-s, 3-s, \dots, s-1}} \frac{\lambda_a + \lambda_b + ub_{rs} + v/b_{rs}}{2} \frac{\lambda_a - \lambda_b + ub_{rs} + v/b_{rs}}{2}. \quad (4.5)$$

The code pairs (u, v) with $(-u, -v)$, evaluating their product as

$$\frac{(4(h_b - h_a) + u^2x + 2uv + v^2/x)^2 - 4\lambda_a^2(u^2x + 2uv + v^2/x)}{16}. \quad (4.6)$$

An unpaired $(0,0)$ contributes $h_b - h_a$. This removes square-root choices from fusion evaluation. The larger-magnitude quadratic root for x avoids a cancelling root. For $r, s \geq 2$ the two roots have product $(1-s^2)/(1-r^2)$; swapping r, s makes the selected root the reciprocal of the other one. Thus (r, s) and (s, r) together supply both finite central-charge poles. For $r = s$ the roots are reciprocal and give one pole. For $s = 1$ one root is zero, so the nonzero root supplies the unique finite pole; there is no additional $r = 1$ contribution. The derivative-times- A product is simplified algebraically

before evaluation: its factors $x - 1$ and $x + 1$ cancel matching factors in the denominator. Unresolved zero denominators are errors, not skipped residues.

Pole geometry and the combined universal residue are cached by null edge, integer weight shift, r, s . Fusion polynomials additionally key the spectator weights and shifts, including their order. These caches last for one ordinary Virasoro block; different outer branching tuples instantiate separate engines.

4.2 The global coefficients

For one normalized Virasoro three-point form, let $\rho(i, j, k) = \rho(L_{-1}^i \nu_1, L_{-1}^j \nu_2, L_{-1}^k \nu_3)$. With $(x)_m$ rising and $x^{\underline{m}}$ falling factorials,

$$\rho(i, j, k) = (h_1 - h_2 - h_3 + i - j + 1 - k)_j \times \sum_{p=0}^{\min(i, k)} \binom{i}{p} (2h_3 + k - 1)^{\underline{p}} k^{\underline{p}} (h_3 + h_2 - h_1)_{k-p} (h_1 + h_2 - h_3 - k + p)_{i-p}. \quad (4.7)$$

The ordinary seed is $\rho(a, b, d)^2 / [a!(2h_1)_a b!(2h_2)_b d!(2h_3)_d]$. The inserted seed is

$$G_{a,b,c,d} = \frac{\rho_{1L3}(a, b, d) \rho_{1R3}(a, c, d) \rho_{R\psi L}(c, 0, b)}{a!(2h_1)_a b!(2h_L)_b c!(2h_R)_c d!(2h_3)_d}. \quad (4.8)$$

Each subscript gives the three ordered weights. The finite sum in (4.7) is independent of j and is cached without that index. Rising/falling products are built incrementally rather than by repeated products inside the sum. Each global norm is cached separately and appears once per internal edge in (4.8).

For the inserted seed, three further caches retain the *complete* vertex factors, including the j -dependent rising product and their assigned norms. The first outer vertex includes the inverse norms of edges 1, 3; the second outer vertex carries no inverse norms; the middle vertex includes those of L, R . Their product is exactly (4.8). Each cache is keyed by its three incident weight shifts and three descendant levels; the external weight is fixed, with zero recursion shift and descendant level. Thus a change on a nonincident edge reuses the entire factor, including its normalization. The cache returns a reference to the stored value. For MPC arithmetic, the final product of these three factors and the incoming amplitude uses one reusable temporary, avoiding repeated allocation of multiprecision mantissas in the assembly sum. All these numerical caches belong to one Virasoro engine and are released when that engine finishes; different branching weights remain separate.

4.3 One forward pass for all coefficients

The transition in (4.1) depends on accumulated internal shifts and the current central charge, but not on the final requested coefficient. Store the amplitude $w_s(c_*)$ of all paths reaching shifted weights $\mathbf{h} + \mathbf{s}$ at central charge c_* . Initialize $w_0(c) = 1$. In increasing total shift, update

$$w_{\mathbf{s} + r\mathbf{s}e_e}(c_{rs}(h_e + s_e)) += R_{rs}^{(e)}(\mathbf{h} + \mathbf{s}) \sum_{c_*} \frac{w_s(c_*)}{c_* - c_{rs}(h_e + s_e)}. \quad (4.9)$$

Keep c_* separate during propagation: summing it away would lose the different outgoing denominators. The common residue is multiplied once after the incoming sum. The reciprocal of each denominator is computed once and stored by the ordered pair of incoming and outgoing central-charge

identifiers; subsequent uses multiply by that reciprocal. The small-denominator guard is applied when the reciprocal is first constructed. This preserves the exact recurrence, although factoring out the residue changes floating-point rounding. After outgoing transitions have been processed, store $\sum_{c_*} w_s(c_*)$ and release that shift's incoming map. Finally,

$$F_{\text{red},\ell}(c; \mathbf{h}) = \sum_{s \leq \ell} \left(\sum_{c_*} w_s(c_*) \right) G_{\ell-s}(\mathbf{h} + \mathbf{s}). \quad (4.10)$$

This is the finite expansion of (4.1), grouped by final shift and terminal global coefficient. Every transition increases a shift by $rs \geq 2$, so a downward-closed finite level set makes the forward pass acyclic. It reuses all internal recursion paths across requested coefficients; it does not reuse a completed block at different external parameters. An exactly zero sum of incoming amplitudes is not used to discard transitions with different c_* . Exact zero residues may be skipped.

4.4 Exact Schottky vacuum

The universal large- c vacuum is computed from CCY Eq. (5.5):

$$F_{\text{vac}} = \prod_{\gamma \in \mathcal{P}} \prod_{m \geq 2} (1 - q_\gamma^m)^{-1/2}. \quad (4.11)$$

Here $\mathcal{P}$ includes primitive conjugacy classes and their inverses separately. The code identifies inverse pairs and uses exponent -1 . For the theta graph the three crossing matrices are

$$\begin{pmatrix} 0 & 1 \\ q_1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & q_2 - 1 \\ 1 & -1 \end{pmatrix}, \quad \begin{pmatrix} 0 & q_3 \\ 1 & 0 \end{pmatrix}. \quad (4.12)$$

Enumerate directed closed walks with no immediate backtracking, modulo cyclic rotation and inversion, excluding proper powers. The direction records the starting sphere, so an odd-length path cannot be mistaken for a closed root. For the projective matrix M_γ of a retained walk,

$$q_\gamma = \sum_{j \geq 1} \frac{1}{j+1} \binom{2j}{j} \left(\frac{\det M_\gamma}{(\text{tr } M_\gamma)^2} \right)^j. \quad (4.13)$$

The trace is a unit at the theta cusp and the leading determinant degree is the walk length. Since the first oscillator is $m = 2$, only walks of length at most $\lfloor N/2 \rfloor$ can contribute through total level N . For the independent-edge cutoff, a walk traversing edge i a total of v_i times can contribute only if $2v_i \leq N$ on every edge, hence its length is at most $\lfloor 3N/2 \rfloor$. After enumerating primitive classes through this length, the code discards any class exceeding an edge-visit bound before computing its multiplier. All multiplier expansions, products, logarithms and exponentials are truncated componentwise in this case. Multiply the factors using their logarithmic series and a truncated exponential, entirely over GMP rationals. This is a cutoff derived from degree, not a separate empirical limit on word length. The inserted sphere collapses by Möbius maps when its light external weight is omitted from this universal factor, giving the same vacuum with $q_2 = \hat{q}_{2,1} \hat{q}_{2,2}$. Both reduced Virasoro factors omit one vacuum, so two copies are restored after physical recovery.

5 Exact target truncation and products

In the physical factor, the inserted field is the identity. Writing G for its Gram matrix, the middle contraction is $G^{-1}GG^{-1} = G^{-1}$ at each level and parity. Consequently the full factorization has the

same physical theta block:

$$\widehat{\mathbb{F}}_f^{(\eta, -\eta)}[\Theta v_{1/2}] = \mathbb{F}_f^{(\eta, -\eta)}(q_1, \hat{q}_{2,1} \hat{q}_{2,2}, q_3) \star_R \mathbb{F}_F[\Theta v_{1/2}]. \quad (5.1)$$

Let Diag retain equal powers of the two split parameters and replace their product by q_2 . Since the physical factor has equal powers,

$$\text{Diag } \widehat{\mathbb{F}}_f^{(\eta, -\eta)}[\Theta v_{1/2}] = \mathbb{F}_f^{(\eta, -\eta)} \star_R \text{Diag } \mathbb{F}_F[\Theta v_{1/2}]. \quad (5.2)$$

Indeed, at a diagonal target (k, k) the physical contribution (j, j) forces the auxiliary contribution to be $(k - j, k - j)$. This proves closure; no off-diagonal physical coefficient must first be computed or divided. The auxiliary diagonal selects precisely the zero-mode action (2.15). Unequal middle levels in each Virasoro factor in (2.17) are nevertheless required on the domain below, before taking the shifted diagonal of their product.

5.1 Total-level cutoff

For a fixed inserted branch define the integer budgets

$$K = \left\lfloor N - 2n_1^2 - (2n_3^2 - 1/8) \right\rfloor, \quad L = K - (2n^2 - 1/8), \quad R = K - (2n'^2 - 1/8). \quad (5.3)$$

Discard branches with $\min(L, R) < 0$. In each individual Virasoro factor keep exactly the downward closure

$$a + b + d \leq L, \quad a + c + d \leq R, \quad a, b, c, d \geq 0. \quad (5.4)$$

Indeed, a lower descendant index extends to an allowed shifted diagonal if and only if $a + d + \max(2n^2 - 1/8 + b, 2n'^2 - 1/8 + c) \leq K$: choose the common final middle power to be this maximum. This criterion is exactly the two inequalities above. For a product target put $t = 0, \dots, \min(L, R)$, choose $a + d \leq t$, and set $(b, c) = (L - t, R - t)$. These are all required targets. For each, sum only the splits of that target between the two Virasoro series. The closure is preserved by every CCY subtraction. Each factor needs unequal middle levels inside it: imposing $b = c$ separately would discard valid terms. The largest allowed middle level is N in the lowest branch, not $2N$.

For ordinary theta sewing, each Virasoro factor instead uses the simplex

$$a + b + d \leq \left\lfloor N - 2n_1^2 - (2n_2^2 - 1/8) - (2n_3^2 - 1/8) \right\rfloor. \quad (5.5)$$

Thus there is a different cutoff for every branching triple in either mode.

5.2 Independent-edge cutoff

For an inserted branch, put

$$L = N - (2n^2 - 1/8), \quad R = N - (2n'^2 - 1/8). \quad (5.6)$$

Each individual Virasoro factor is evaluated at nonnegative integer descendant levels in the box

$$\begin{aligned} 0 \leq a \leq \lfloor N - 2n_1^2 \rfloor, \quad 0 \leq d \leq N - (2n_3^2 - 1/8), \\ 0 \leq b \leq L, \quad 0 \leq c \leq R. \end{aligned} \quad (5.7)$$

Discard a branch if any upper bound is negative. In the product of the two factors retain the same first- and third-edge bounds and impose

$$2n^2 - 1/8 + b = 2n'^2 - 1/8 + c \leq N. \quad (5.8)$$

Here a, b, c, d in the product are the sums of the two factors' descendant indices. Equation (5.7) is precisely the downward closure of these shifted-diagonal targets: any index in the box can be completed by increasing its smaller middle total to the larger one, leaving its first and third levels unchanged. Conversely every lower index of an allowed target lies in the box. Thus individual factors need unequal middle levels, but neither middle cutoff is doubled. Product assembly sums only the splittings of the required targets.

For ordinary sewing the corresponding three-dimensional box is

$$0 \leq a \leq \lfloor N - 2n_1^2 \rfloor, \quad 0 \leq b \leq N - (2n_2^2 - 1/8), \quad 0 \leq d \leq N - (2n_3^2 - 1/8). \quad (5.9)$$

Every stage uses these componentwise bounds, including branching requests, CCY, products, auxiliary sewing, division and vacuum restoration. The code does not first compute a total-level- $3N$ block and then discard coefficients. The physical box has $(2N + 1)(N + 1)^2$ monomials: 2,541 at $N = 10$, compared with 506 at total level 10. Both truncations are downward closed, so the forward recursion and triangular recovery close on their requested sets.

5.3 Reuse under middle-edge transposition

BPZ transposition at the insertion point one gives

$$\begin{aligned} F(h_1, h_L, h_R, h_3; h_\psi; c \mid q_1, q_L, q_R, q_3) \\ = F(h_1, h_R, h_L, h_3; h_\psi; c \mid q_1, q_R, q_L, q_3). \end{aligned} \quad (5.10)$$

The code computes each product once for an unordered pair $\{n, n'\}$ at fixed n_1, n_3 , then transposes its middle exponents for the reverse orientation. Both orientations retain their own branching coefficients, norms, and parity factors. The stored product is released after its second use.

6 Direct fermion factor and physical recovery

Use distinct descending positive half-integer modes $\mathbf{A}$ on the NS edge and distinct descending positive integer modes $\mathbf{B}, \mathbf{C}$ on Ramond edges. Include $u^{\mathbf{b}}, u^{\mathbf{c}}$ with $\mathbf{b}, \mathbf{c} \in \{0, 1\}$. Put $\delta_1 = \#\mathbf{A}$, $\delta_2 = \#\mathbf{B} + \mathbf{b}$, $\delta_3 = \#\mathbf{C} + \mathbf{c}$ modulo two. Only $\delta_1 + \delta_2 + \delta_3 = 0$ contributes. In the following state sum, use 1 for ordinary sewing and $Q(-1)^{\mathbf{b}}/\sqrt{2}$ for the inserted diagonal:

$$\begin{aligned} \sum_{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{b}, \mathbf{c}} q_1^{|\mathbf{A}|} q_2^{|\mathbf{B}|} q_3^{|\mathbf{C}|} \eta_1^{\delta_1} \eta_2^{\delta_2} \eta_3^{\delta_3} (-1)^{\delta_1 + \delta_1 \delta_2 + \delta_1 \delta_3 + \delta_2 \delta_3} \\ \times \rho_F(\Psi_{-\mathbf{A}} \mathbf{1}, \Psi_{-\mathbf{B}} u^{\mathbf{b}}, \Psi_{-\mathbf{C}} u^{\mathbf{c}})^2 \begin{cases} 1, & \text{ordinary,} \\ Q(-1)^{\mathbf{b}}/\sqrt{2}, & \text{inserted diagonal.} \end{cases} \end{aligned} \quad (6.1)$$

Each Fock norm is $(-1)^{\delta_i}$. In ordinary sewing their inverse product is one on an allowed even-parity triple. In inserted diagonal sewing the additional edge-2 inverse norm cancels the BPZ norm in the middle matrix element, leaving $Q(-1)^{\mathbf{b}}/\sqrt{2}$. For a total cutoff retain $|\mathbf{A}| + |\mathbf{B}| + |\mathbf{C}| \leq N$; for an independent-edge cutoff bound each of $|\mathbf{A}|, |\mathbf{B}|, |\mathbf{C}|$ separately by N . The constant terms are $1 + \eta_2 \eta_3$ and $Q(1 - \eta_2 \eta_3)/\sqrt{2}$, respectively.

Three-point forms are evaluated by the fermion Ward identity. Remove the largest mode in the first nonempty slot and move its contour using, for target NS mode $k - 1/2$ or Ramond mode n , respectively,

$$z^{k-1/2}(z-1)^{-1/2}, \quad z^{-1/2}(z-1)^{-n-1/2}, \quad z^{-n-1/2}(z-1)^{-1/2}. \quad (6.2)$$

The local expansions give half-integer binomial coefficients, frame phases $(i, 1, -i)$, and graded contour signs. Solve for the original form; other terms have lower nonzero-mode level. Stop each local expansion when its modes annihilate the states, with a bound computed from the largest occupied mode. Store every resulting three-point form across the entire sewing sum.

For exact arithmetic temporarily rescale $u^1 \rightarrow \sqrt{2}u^1$ and strip the phase i^{δ_2} . The stored rational form satisfies

$$\rho_F = i^{\delta_2} 2^{-(b+c)/2} \rho_F^{\text{rat}}. \quad (6.3)$$

Its ground values are 1, 2, and its zero-mode matrix entries are $1/2, 1$. All Ward arithmetic and sewing are then rational. Restore only the common $Q/\sqrt{2}$ for the inserted factor at the chosen numerical precision. Only diagonal zero-mode transitions are enumerated in production.

For either sector, subtract positive-level auxiliary convolutions using already solved lower coefficients. Divide the remainder by 2 for ordinary sewing or $\sqrt{2}Q$ for inserted diagonal sewing. Explicitly, for any physical monomial q^k and the appropriate numerator/auxiliary pair,

$$\mathbb{F}[q^k] = \frac{\widehat{\mathbb{F}}[q^k] - \sum_{\substack{0 \leq r \leq k \\ r \neq 0}} \mathbb{F}_F[q^r] \star_R \mathbb{F}[q^{k-r}]}{\begin{cases} 2, & \eta\eta' = 1, \\ \sqrt{2}Q, & \eta\eta' = -1. \end{cases}} \quad (6.4)$$

The numerator on the negative side means its diagonal projection. With the reduced Virasoro sum as input, this recurrence first gives $\mathbb{F}/F_{\text{vac}}^2$; multiply by F_{vac}^2 afterwards. The sector identity (1.5) proves uniqueness inductively because the auxiliary constant acts as the displayed nonzero scalar on that sector. Equivalently its sector inverse has identity $(1 + \eta\eta'\eta_2\eta_3)/2$, rather than the identity of the full algebra. The auxiliary higher coefficients preserve the corresponding sector, so each exact remainder stays there.

The implementation orders monomials by total degree, stores positive-degree auxiliary terms in that order, and updates future remainders immediately after solving a coefficient. Updates stop at the chosen total or componentwise cutoff and never revisit a solved coefficient. Only physical triples and their eight parity slots are stored.

7 Precision, timings, and PBW comparison

Machine arithmetic uses `std::complex<double>`; higher precision uses an owning MPC type with MPFR/GMP and the same templated mathematical routines. At 40 digits it uses 136 bits. This does not mean 40 certified output digits: the mode-action solves still use double-precision column-rank selection and LU preconditioning. Outer branching instead uses the stored recursion of Section 3.2, with local arithmetic at working precision. At decimal precision D , the selected-system residual target is $10^{-\max(20, D-15)}$ and the full-system residual guard is $10^{-\max(15, D-20)}$. Sparse pruning uses $10^{-\max(20, D-20)}$. Native action residuals are required below 10^{-8} , and native sparse pruning uses 10^{-13} . Each outer Ward residual is the absolute value of the sum of its terms divided by one plus the sum of their absolute values; its guard is 10^{-8} at machine precision and $10^{-\max(15, D-20)}$ at D digits. At most 30 refinement iterations are attempted. Ill-conditioned rank selection or failed convergence raises an error rather than silently changing the basis.

The sector residual is evaluated during recovery, before vacuum restoration, on the coefficients of $\mathbb{F}/F_{\text{vac}}^2$. It is the distance of a parity vector from $[v + \eta\eta'(\eta_2\eta_3) \star_R v]/2$, divided by the largest of one and its component magnitudes. The default policy stops above 10^{-8} . It is not recomputed on the

final restored arrays. Ordinary default recovery projects accepted small residuals; inserted default recovery retains its coefficients. The explicit `record` policy retains unprojected coefficients in both modes and records the residual. Timing runs below use this policy, so completion alone is not an accuracy certificate.

7.1 Fresh independent-edge level-10 measurements

Both production pipelines use the stored outer recurrence. The parameters are

$$b = 7/5, \quad (P_1, P_2, P_3) = (11/23, 13/29, 17/31), \quad p_1 = f = 0, \quad (\eta, \eta') = (+, +) \text{ or } (+, -). \quad (7.1)$$

Each sector and precision is a separate process with empty numerical caches. Runs are sequential; within-process reuse remains enabled. The build uses Apple clang 17 with `-O3` on macOS arm64 with Accelerate. Fermion Ward sums and Schottky series use exact GMP rationals. The internal timer includes branching, CCY, products, assembly, direct fermion, convolution, and Schottky restoration. A parent timer additionally includes process startup, final output serialization, and exit. Compilation is excluded. The executable and source hashes, commands, diagnostics, stage times, and full physical coefficient arrays are saved under `C++/results/per_edge_level10_2026-09-11/optimized_full_pipeline_40dps/`.

These runs impose $N = 10$ independently on each edge, retaining all 2,541 monomials and 20,328 parity components in each sector. They use 40 decimal digits of working precision (136 bits), with the fermion Ward sums and Schottky series evaluated exactly. All table entries are seconds.

Stage	$\eta\eta' = 1$	$\eta\eta' = -1$
Initial mode actions	9.497	9.594
Outer branching recurrence	0.441	0.450
Middle recurrence	0.000	0.013
Reduced CCY	296.593	1460.243
Virasoro products	26.759	88.559
Branch assembly	0.356	0.577
Direct fermion	5.220	5.221
Sector division	3.115	3.130
Schottky vacuum and square	72.379	72.492
Vacuum restoration	0.375	0.380
Other pipeline overhead	1.801	14.620
Internal total	416.537	1655.279
Startup, serialization and exit	0.081	0.081
Full wall time	416.619	1655.360

The full wall times are **6 min 57 s** and **27 min 35 s**. Other pipeline overhead is the difference between the independently timed internal total and the named stages; it includes work and cache cleanup outside their timers. Branching, including initial actions but excluding the middle recurrence, takes 9.938 s and 10.044 s. The outer Ward recurrence itself takes 0.441 s and 0.450 s.

Recorded work	$\eta\eta' = 1$	$\eta\eta' = -1$
Direct boundary values	96	192
Stored recursive branching values	1,704	3,408
Assembled branch tuples	899	1,591
Virasoro blocks	1,798	1,620
Reused transposed products	0	781
Global-seed summands	42,624,112	1,740,820,224
CCY transitions	51,242,176	839,918,556

The ordinary wall time was 508.004 s before the latest CCY reuse changes; its CCY stage decreased from 390.230 s to 296.593 s. The former inserted run was interrupted and had projected to about 4.5 hours. That forecast is not a completed baseline measurement. Interrupted and overlapping attempts are excluded from the tables above.

Directed checks and accuracy limits

Both saved outputs contain the complete requested box and only finite physical components. Source and executable hashes match the launch records. The maximum recorded recovery-sector residuals are 2.226×10^{-15} and 8.837×10^{-16} , respectively. These test sector consistency before vacuum restoration, not independent physical-coefficient accuracy.

Comparing all 20,328 ordinary components with the saved pre-change 40-digit result gives maximum scaled difference 2.107×10^{-20} and absolute difference 8.907×10^{-9} ; coefficients reach 4.166×10^{12} . Here the scale is $\max(1, |a|, |b|)$. This checks the implementation change at the same working precision. The full inserted box has no completed pre-change or independent reference.

At independent cutoff 2, both physical pipelines were compared with saved results on all 45 monomials; after the CCY optimization their scaled differences from the saved pre-change box are below 2.97×10^{-35} . Individual inserted Virasoro factors at edge cutoffs 6 and 8 agree with the former implementation within 8.88×10^{-32} in scaled difference. Exact checks compare small Schottky and fermion boxes with projections of their total-degree series and check the shifted-diagonal product on an asymmetric domain. No additional physical PBW block was computed.

The directed optimization evidence is saved under `C++/results/ccy_vertex_reuse_2026-09-11/`; output inspection and cutoff validation are in `C++/results/per_edge_level10_2026-09-11/`.

7.2 PBW workload estimate, without running PBW

PBW times below refer to the existing physical SCA implementation in Python, with its numerical backend stated explicitly; they are not estimates for a hypothetical optimized C++ PBW code. The dimensions used for work counts are

$$\sum_{a \geq 0} d_{\text{NS}}(a) x^a = \prod_{m \geq 1} \frac{1 + x^{2m-1}}{1 - x^{2m}}, \quad \sum_{l \geq 0} d_{\text{R}}(l) y^l = \prod_{m \geq 1} \frac{1 + y^m}{1 - y^m}. \quad (7.2)$$

The NS index is twice the level and the Ramond dimension is per parity. Counts of required tensor entries, reused Gram entries, and contractions are computed from these generating functions and the actual PBW loop bounds. The cost model uses saved physical-block logs and a short arithmetic-only calibration of their backend at 136 and 384 bits. The sign choice does not change basis sizes; cached reuse of the same three-point form for identical vertices must be accounted for separately when

comparing the two sectors. A fitted range measures sensitivity to the cost model, not a confidence interval or a guaranteed runtime bound.

For the independent-edge level-10 box, the required PBW work is

Quantity	Count
Physical monomials	2,541
Largest NS Gram dimension	161
Largest R Gram dimension, per parity	232
Entries in distinct Gram matrices	450,469
Cubic Gram sum (inversion-cost proxy)	78,765,711
Requested vertex entries, negative sector	1,318,075,412
Dense contraction multiplications	254,809,434,024
Largest single vertex tensor	8,665,664 entries

The per-edge Gram dimensions are unchanged from total level 10, but the box includes many more combinations of high levels. For dimensions d_1, d_2, d_3 , two vertices and the two parity sectors request $4d_1d_2d_3$ form entries; successive Gram contractions and the final pairing use $2d_1d_2d_3(d_1 + d_2 + d_3 + 1)$ complex multiplications. The sum of dimension cubes is a dense-inversion work proxy, not an exact operation count. Identical vertices in the positive sector can share form evaluations; only the fitted form-entry cost is allowed to fall by a factor between one half and one. Gram and contraction costs remain.

136-bit PBW model	$\eta\eta' = 1$	$\eta\eta' = -1$
Lean	28.8–63.4 h	57.6–63.4 h
Full	97.3–196.0 h	119.3–196.0 h

These are estimates for the existing Python/FLINT PBW implementation. Saved 384-bit PBW timing fits are rescaled using synthetic scalar and matrix operations in its actual arithmetic wrappers at 136 and 384 bits. The arithmetic-only calibration took 1.30 s and evaluated no physical block, Gram matrix or Ward identity. Measured 136/384-bit time ratios were about 0.91–0.92 for scalar operations and 0.51–0.73 for matrix products. The estimates allow unchanged Python/cache overhead.

The lean fit omits explicit cubic-Gram and contraction features; the full fit includes them. Their spread exposes model sensitivity, rather than forming a confidence interval. Larger matrix shapes, internal Ward work and memory pressure may change these costs; sufficient memory is assumed. The C++ and PBW comparison now uses the same 136-bit working precision, but different implementations (MPC versus Python/FLINT). No full 136-bit PBW block was timed for this box.

Integer counts, saved-fit provenance and the arithmetic calibration are in
C++/results/per_edge_level10_2026-09-11/work_estimates.json and
C++/results/per_edge_level10_2026-09-11/pbw_40digit_arithmetic_calibration.json.

7.3 Reproduction and evidence

Under `C++/include/ramond/`, the stages are separated into `free_field.hpp` and `branching.hpp` (actions and branching), `ccy.hpp` (reduced Virasoro factors), `schottky.hpp` (vacuum), `fermion.hpp` (direct auxiliary sewing), and `pipeline.hpp` (branch-specific products, assembly, and recovery). Partition lists, solved action data, outer primary coefficients, and auxiliary Ward values are reused within a run. Pole, fusion, inverse-denominator, global-core and complete inserted-vertex caches belong to their individual Virasoro engine; distinct branching weights do not share those numerical values. This distinction fixes the meaning of fresh runs and reuse in the timing tables.

From the repository root:

```
make -C 'C++' -j2
python3 'C++/tools/time_current_pipelines.py' --levels 10 --dps 40 \
  --truncation per-edge --output '/tmp/ramond_per_edge_level10'
```

The coefficient convention in JSON is $[a, l, l, d]$ for $q_1^{a/2} q_2^l q_3^{d/2}$, with even d . Under the independent-edge cutoff, $0 \leq a, d \leq 2N$ and $0 \leq l \leq N$; under a total cutoff, $a + 2l + d \leq 2N$. The JSON identifies the selected truncation and records the independent limits as `q_level_cutoffs`. Parity slot is $\epsilon_1 + 2\epsilon_2 + 4\epsilon_3$. Directed checks of the stored recurrence cover all 768 ordinary-support and 880 inserted-support outer coefficients through level 5. Agreement with the saved reference coefficients is within 3.76×10^{-25} in scaled difference at 40 digits, including equal Ramond momenta. The equal-momentum case exercises vanishing first prefactors; each low-level run checks 608 second Ward identities across both vertex signs and Ramond parities. The independent-edge output checks and their limits are reported above. All reference physical blocks were already stored; no new physical PBW block was computed. Regenerate the timing and PBW tables from saved records, without evaluating blocks, using

```
python3 'C++/tools/render_per_edge_notes.py'
```

References

- [1] M. Cho, S. Collier and X. Yin, *Recursive Representations of Arbitrary Virasoro Conformal Blocks*, [arXiv:1703.09805](#).